

Mini Hackathon: Blocking for Entity Matching

Overview

In this mini hackathon, your goal is to design a blocking algorithm for an entity matching task. You will be given with the files `TableA.csv` and `TableB.csv`, which contain product information. You are also provided with a set of 100 matching pairs, `100_matches.pkl`, which may be used for model training, feature engineering, or whatever you like. **these 100 pairs must not appear in your final candidate set.**

Task Description

Your task is to produce a list of candidate matching pairs of the form (same format as `100_matches.pkl`):

$$\{(id_a, id_b), \dots\}$$

where `id_a` is a row identifier from `TableA`, and `id_b` is a row identifier from `TableB` (just as we saw in the tutorial). The set should contain at most **2000 pairs**. Your goal is to design a blocking strategy that retains high recall while restricted to the 2000 pairs limitation.

Submission Instructions

You must submit a zip file named `<sid>.zip` (where `<sid>` is your ID.) containing the following :

1. A Python script or Jupyter notebook with your full code. The code does not need to be polished or well-organized, but it must include all relevant logic.
2. A file named `sid.pkl`, where `sid` is your ID. This file must contain your candidate set as a Python set of `(id_a, id_b)` tuples.

Helper Functions

We provide a Python script containing two helper functions:

- `validate_candidate_set(candidate_set, tableA_ids, tableB_ids)`: Ensures your submission is valid (e.g., correct format, maximum size, valid IDs).
- `save_submission(candidate_set, sid)`: Saves your final candidate set as a `sid.pkl` file, where `sid` is your ID.

Download or copy the contents of the file `helpers.py` and paste the functions to your code. **Do not change the parameter value of `max_pairs` in `validate_candidate_set`.**

Make sure you call the validation function before saving your result. For example:

```
validate_candidate_set(my_candidate_set, set(tableA['id']), set(tableB['id']))
save_submission(my_candidate_set, '123456789')
```

Grading Scheme

Your final grade will be based on the **recall** (i.e., *pair completeness*) of your blocking output with respect to the hidden ground truth.

- If your recall is below 0.6, your grade will be:

$$\text{Grade} = \text{Recall} \times 150$$

- If your recall is 0.6 or above, your grade will be determined by:

$$\text{Grade} = \max\{100 - (\mathbf{r}\text{-}\mathbf{ranking}), 90\}$$

where **r-ranking** is your rank relative to other students (e.g., rank 0 is the best, rank 1 is second-best, etc.).

Important: Your recall will be computed using a hidden set of test pairs (taken from the two given tables). We will only test whether your candidate set includes these correct matches.

Enjoy!